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1. Purpose

The objective of this design verification test protocol is to define the test method and objective evidence needed to verify the synchronization accuracy of surgical navigation systems, comprising Orbit base stations and Axon SIUs, as defined in the referenced functional specifications.

2. Background

Synchronization between the base station and SIUs is critical to accurate tracking. In this test, synchronization is measured at the hardware level, both to measure synchronization accuracy and the maximum error after loss of synchronization.

The SIU will report a fault after loss of synchronization for 84 ms. Synchronization compensates for part-to-part variations in the Base Station and SIU. For example, the individual clocks alone can cause a 40 ppm error, so after 84 ms, the clocks could differ by up to 3.36 μ s. An error less than 50 μ s is negligible for position accuracy of the overall Ommo system.

3. Scope

This test applies to the following components.

- 3.1. 10001 Orbit Base Station
- 3.2. 12186 Single Port SIU
- 3.3. 13469 3 Port SIU, 1.0m
- 3.4. 13005 3 Port SIU, 3.0m

4. References

- 4.1. DHF-100140: Orbit DIOVV Matrix

5. Equipment and Materials

- 5.1. Computer with at least three USB ports
- 5.2. Ommo Firmware Updater - Internal (Ommo Hardware Test App)
- 5.3. CNC Test App (installed on computer)
- 5.4. 11658 OMMO USB-C TO USB-A CABLE, 5M
- 5.5. 11918_B USB Programming and Test Breakout Board
- 5.6. 12246 USB TEST BOARD
- 5.7. Oscilloscope
- 5.8. Any sensor, such as 12252 Single Mag Sensor, 1.5m

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6. Test Setup

6.1. This test must be performed outside of wireless range of base stations other than the DUT. Otherwise, the SIU may synchronize with the non-DUT base station.

6.2. Install test firmware on each Base Station DUT

6.2.1. Open Ommo Firmware Updater app

6.2.2. Navigate to the “Master Firmware Updater” tab

6.2.3. For each Base Station DUT

6.2.3.1. Click “Scan Hardware” if necessary

6.2.3.2. Select a filename that contains “synch_test” from the Base Station dropdown menu

6.2.3.3. Click “Update” button and wait for “Firmware update completed successfully” message.

6.3. Install test firmware on each SIU DUT

6.3.1. Open Ommo Firmware Updater app

6.3.2. Navigate to “Master Firmware Updater” tab

6.3.3. For each SIU DUT

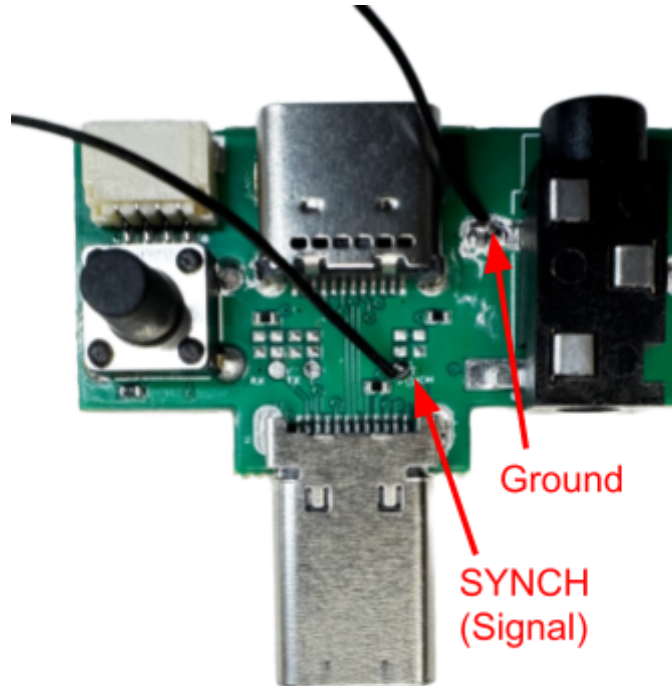
6.3.3.1. Click “Scan Hardware” if necessary

6.3.3.2. Select filename that contains “synch_test” from the SIU dropdown menu

6.3.3.3. Click “Update” button and wait for “Firmware update completed successfully” message.

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6.4. 11918, USB Programming and Test Breakout Board

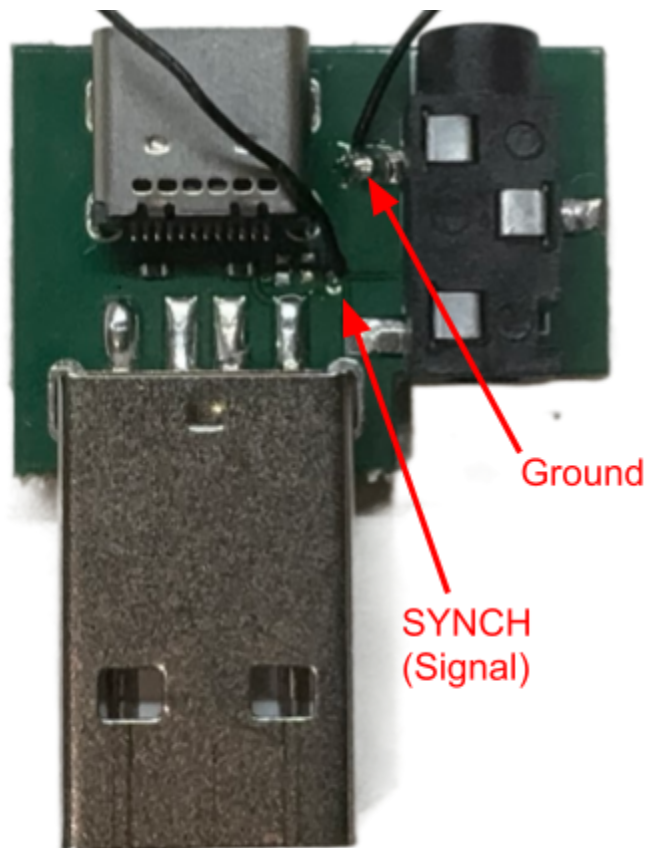


6.4.1. Depopulate: R4, R5

6.4.2. Add lead wires to the SYNCH and ground

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6.5. 12246 Test Board



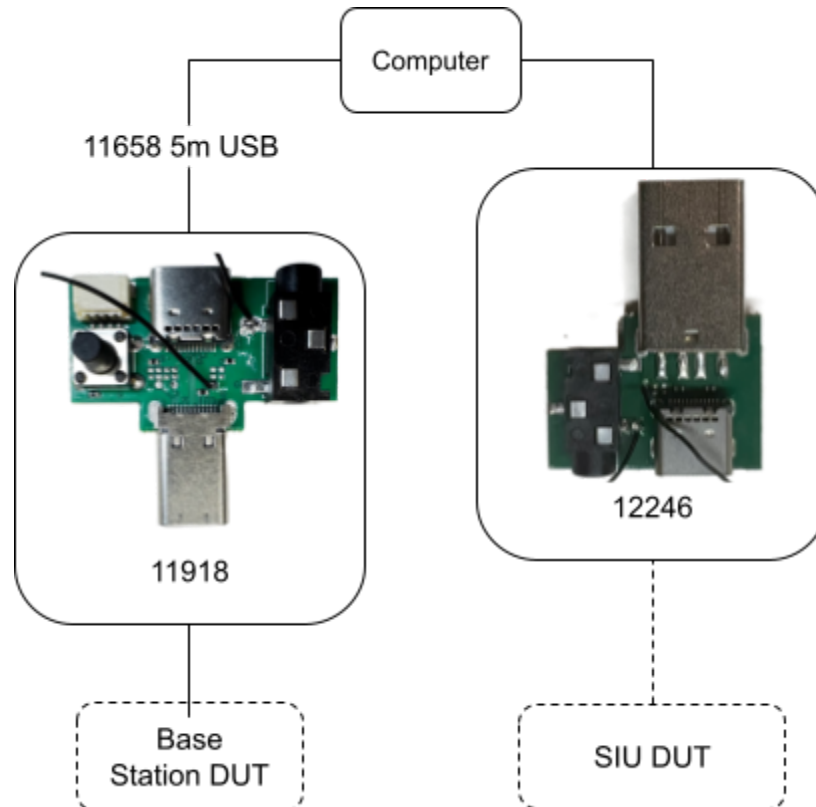
6.5.1. Depopulate R5

6.5.2. Add lead wires to the “Synch” pin and ground

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6.6. Connect the computer and breakout boards as shown in the figure

- 6.6.1. Connect the breakout board (11918) to the computer (USB-C). Base Stations (DUT) will be connected to the breakout board with 5m USB cable (11658, dotted lines) in Test Method.
- 6.6.2. Connect the test board (12246) to the computer (USB-A). SIUs (DUT) will be connected to the test board via integrated SIU cable (dotted lines).



6.7. Connect a sensor to the SIU DUT

6.8. Connect and configure oscilloscope

- 6.8.1. Connect one channel of oscilloscope to lead wires on Breakout Board (11918, Base Station) and connect second channel of oscilloscope to lead wires on Test Board (12246, SIU).
- 6.8.2. Set voltage scale to 1V per division for base station signal and 50mV per division for SIU signal.

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7. Sample Size

7.1. To verify with 90% confidence and 90% reliability that Axon systems meet the performance parameters defined in acceptance criteria, the required sample size n is determined using the binomial test formula assuming zero observed failures:

$$R^n = 1 - C$$

Where:

- C = confidence level = 90%
- R = reliability = 90%

Rearranging to solve for n :

$$n = \frac{\ln(1-C)}{\ln(R)} = \frac{\ln(1-0.90)}{\ln(0.90)} \approx 21.8$$

Therefore, a minimum of 22 base station-SIU systems must be tested with zero failures to demonstrate, with 90% confidence, that at least 90% of devices meet the performance parameters in design inputs listed in Section 10 (Acceptance Criteria).

7.2. SIUs 13469 (3 Port SIU, 1.0m) and 13005 (3 Port SIU, 3.0m) differ only in cable length, which does not impact measurement performance. Therefore, results obtained from testing one device are representative of both, and only one device requires verification testing.

7.3. In summary, 22 samples of each device in unique pairs shall be tested

7.3.1. 22 pairs: 22 x 10001 Orbit Base Station and 22 x 12186 Single Port SIU

7.3.2. 22 pairs: 22 x 10001 Orbit Base Station and 22 x 3 Port SIU (13469 or 13005)

Note: the same 22 base station samples can be used in both sets

8. Test Method

For each Orbit Base Station and SIU pair, complete the following steps:

8.1. Connect DUT

8.1.1. Connect base station (DUT) to Breakout Board (11918) via 5m USB cable (11658).

8.1.2. Connect SIU (DUT) to Test Board (12246) via the integrated USB cable.

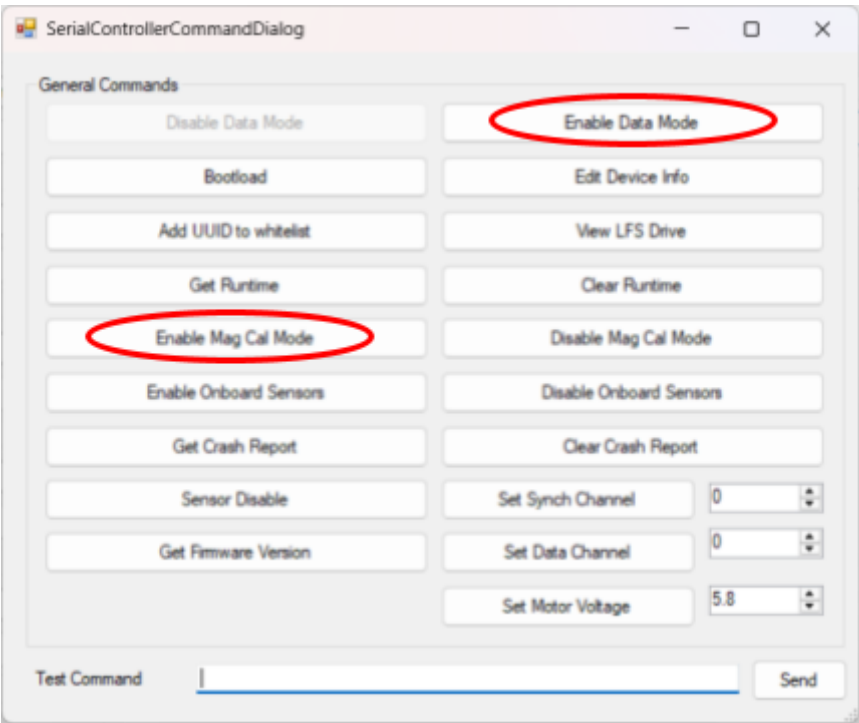
8.1.3. Record DUT serial numbers in the data sheet.

8.2. Open and configure CNC Test App

8.2.1. Select Base Station and click "Open"



8.2.2. Click “Command”, then click “Enable Mag Cal Mode” and “Enable Data Mode” in the SerialControllerCommnadDialog window. Close the SerialControllerCommnadDialog window.

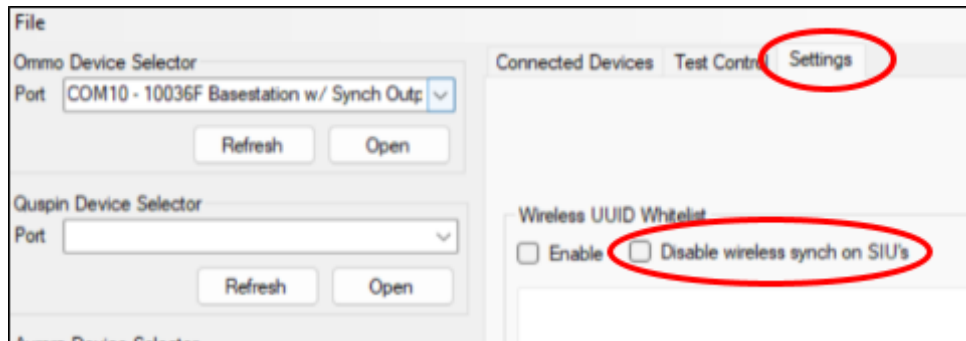


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8.2.3. Navigate to the Settings tab and make sure the “Disable wireless synch on SIU’s” box is not checked. Return to Connected Devices tab.

8.2.3.1. Listen or feel to confirm that the Base Station is rotating.

8.2.3.2. Confirm that the PPS field is displaying 3003 (+/- 10).



Note: It is not necessary to open the SIU port.

8.3. Measure synchronization accuracy

8.3.1. Set oscilloscope persistence to 500 ms.

8.3.2. Set oscilloscope to trigger on the rising edge of the Breakout Board (Base Station) signal

8.3.3. Adjust the oscilloscope display so that the inflection points of the Base Station and SIU synchronization signals are clearly visible and occupy the majority of the display. If the SIU signal is falling at the rising edges of the Base Station signal, change the oscilloscope to trigger on the falling edge of the Base Station signal. Use the oscilloscope cursors to measure the Synchronization Accuracy, i.e., the lag between the Base Station and SIU synchronization signals, and record it in the datasheet.

The following images show examples of both a rising edge and falling edge synchronization signal. The oscilloscope trigger should be adjusted to permit measuring a rising edge of the SIU signal. SIU jitter is visible due to persistence setting (see Section 6). Cursors are used to identify the inflection points of the SIU (solid vertical white line) and Base Station (dashed vertical white line) signals. The SIU inflection point that is farthest from the Base Station inflection point within the observed jitter is selected. The oscilloscope calculates the Synchronization Accuracy, i.e., inflection point lag (“BX-AX = 108ns” in this first example).

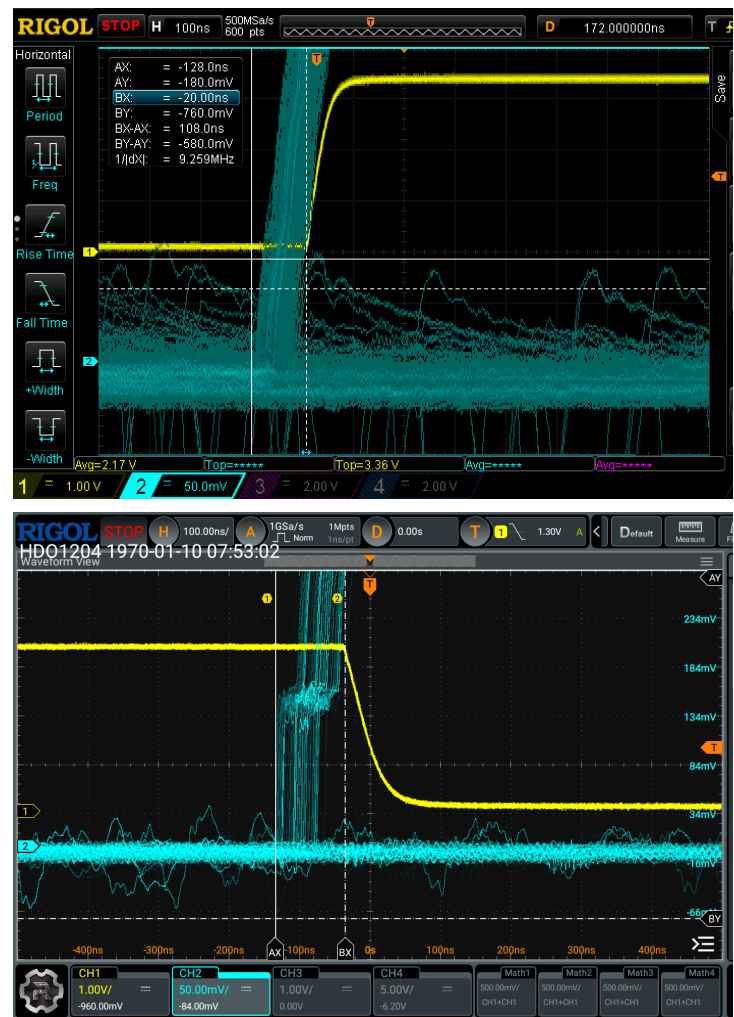


Figure: Examples of Base Station (yellow) and SIU (blue-green) synchronization signals.

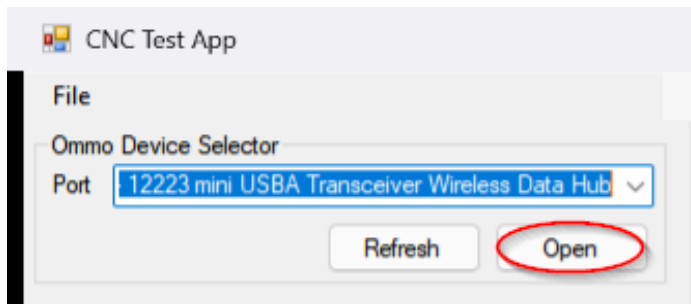
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8.4. Measure clock drift after loss of synchronization

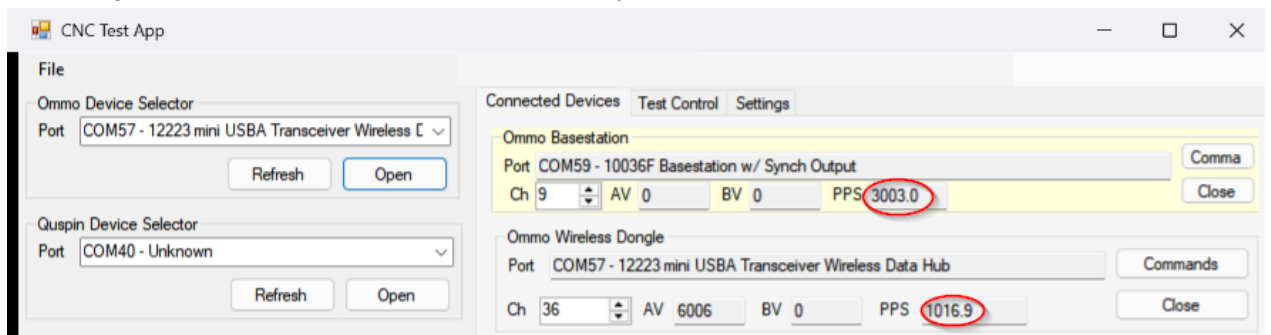
8.4.1. Turn off oscilloscope persistence

8.4.2. Set oscilloscope to trigger on both edges of the Test Board (SIU) signal with 5ms timeout

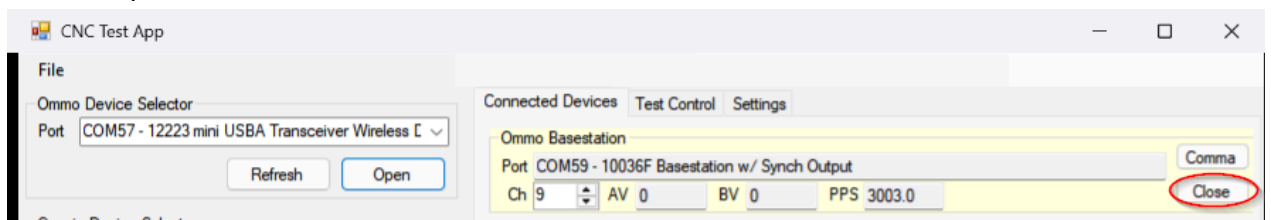
8.4.3. In CNC Test App, select the 12223 Wireless Receiver and click “Open”

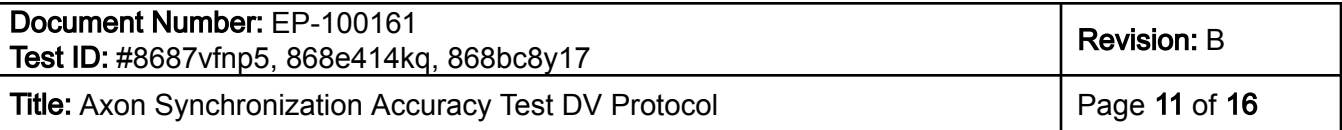


8.4.4. Observe that the Base Station reports ~3003 PPS, and Wireless Receiver reports ~1001 PPS, indicating that the Base Station and SIU have synchronized



8.4.5. In CNC Test App, close the connection to the base station. This stops wireless transmission from the Base Station. ~80 ms later, the SIU will recognize the loss of signal from the Base Station, and the SIU will stop outputting the synch signal. The loss of signal should trigger the oscilloscope.

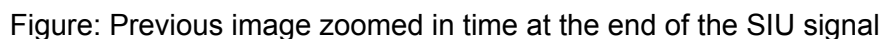




In the following image, you can see the SIU signal has a 1ms period. The dashed cursor is at the end of the last full 1ms period. The solid cursor is 84ms before that point, approximately where the base station stopped transmitting.



The following oscilloscope image is zoomed in on the final 1ms period of the SIU signal, near the dashed line in the previous image. The solid cursor is at the end of the final 1ms period of the SIU signal. The dash cursor is at the point where the firmware has reacted to the loss of synchronization and stopped the SIU signal. Note that in this example, the final 1ms period trailing edge is rising, **but it could also be falling**. Also note that if the firmware reacts while the SIU signal is already low, there will not be a final “false edge”.



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The last image in this step shows the solid cursor is at the final edge of the final 1ms period of the SIU signal, and the dashed cursor is at the corresponding edge of the Base Station signal. **Note that in this example, the edges are both rising, but the edges could be either direction.** Also note that in this example, the SIU edge leads the Base Station edge, but it is also possible for the Base Station edge to lead the SIU edge. In any case, measure the time between the edges, which is the maximum clock drift after synchronization loss. In this example, the clock drift is 2.68 μ s

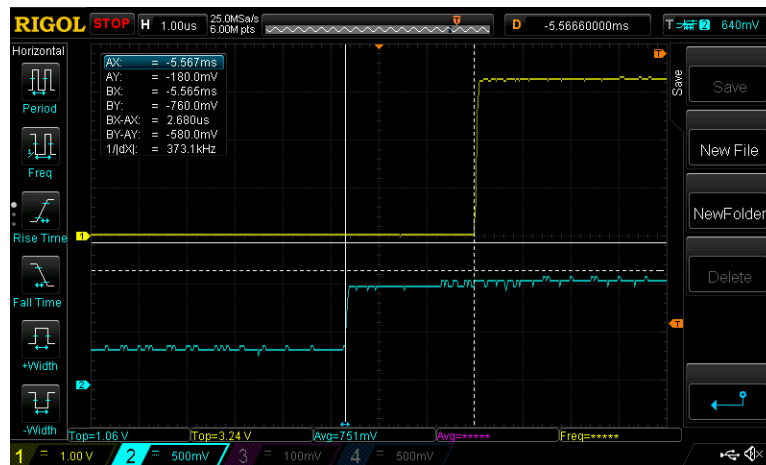


Figure: Previous images further zoomed in on the trailing edge of the final 1ms period of the SIU

Confirm in the CNC Test App console window that the SIU sent a fault to the Wireless Receiver. The AlertSynchFail message is the fault notification. The number “21” indicates the number of missing synchronization messages before transmitting the fault. Synchronization messages are transmitted by the Base Station every 4ms; $21 * 4 \text{ ms} = 84 \text{ ms}$.

```

C:\workspace\cnc_testapp\cnc_testapp x + -
00387, "v1": -0.0002535, "z1": -7.5E-05, "v2": -0.000387, "v2": -0.0002535, "z2": -7.5E-05 } ] ], "icDataFormat": "IC_
DATA_FORMAT_MNC5983_RESET", "nominalPoseVecTuSens": [ -0.009147, 0.002649, 0.002761, 0.707, 0, 0, 0.707 ], "trackingUnit
Index": 1 ] ], "devicePartNum": 85, "calibrationDate": "1728357418", "uid": "5756756839118362899", "dataSourcingStrat":
"055_FIRMWARE", "deviceInfoUser": { "userDeviceType": 255 } } ], "timestampTicksPerSecond": 16000000 } ], "portStatus
List": [ "PORT_STATUS_ERROR", "PORT_STATUS_CONNECTED", "PORT_STATUS_DISCONNECTED", "PORT_STATUS_DISCONNECTED" ] }
Wireless data packets will contain 19 bytes, and take 112 us to transfer
638912203520150820: Sending command packet: UsbSetGoodIdBitfield: 01 00-00-00
638912203520160807: Received command packet: Ack: Success
638912203520160807: Sending command packet: UsbDirectCommWirelessRelayPacket: 40-00-1A-00-00-00
638912203520160807: Received command packet: Ack: Success
6389122035201716409: Received command packet: UsbDirectCommDisconnected:
638912206039281632: Received command packet: UsbDataModeWirelessRelayPacket: SDCommand: AlertSynchFail ID: 0 Payload: 4F
65D86000000015
Command Received in dongle: CNCTestApplication.omoCommandResponse
Warning synch failure reported from : 0 count: 21
638912206039281632: Received command packet: UsbDataModeWirelessRelayPacket: SDCommand: AlertSynchFail ID: 0 Payload: 4F
65D86000000017
Command Received in dongle: CNCTestApplication.omoCommandResponse
Warning synch failure reported from : 0 count: 23
638912206039431557: Received command packet: UsbDataModeWirelessRelayPacket: SDCommand: AlertSynchFail ID: 0 Payload: 4F
65D86000000018
Command Received in dongle: CNCTestApplication.omoCommandResponse
Warning synch failure reported from : 0 count: 24
638912206039431557: Received command packet: UsbDataModeWirelessRelayPacket: SDCommand: AlertSynchFail ID: 0 Payload: 4F
65D86000000019
Command Received in dongle: CNCTestApplication.omoCommandResponse
Warning synch failure reported from : 0 count: 25
638912206039441564: Received command packet: UsbDataModeWirelessRelayPacket: SDCommand: AlertSynchFail ID: 0 Payload: 4F
65D8600000001A
Command Received in dongle: CNCTestApplication.omoCommandResponse

```

Figure: Console

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8.4.7. Analysis

To establish verification of design inputs 8685pr2yd and 8685wxruv, the synchronization signal accuracy of all DUTs must be less than 4 μ s.

To establish verification of design inputs 868dt23nr and 868dt22ym, the clock drift must be less than 50 μ s.

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9. Acceptance Criteria

Parameter	Design Input	Acceptance Criteria
Base Station Wireless Synchronization	Requirement ID: 8685pr2yd A Base Station shall broadcast a wireless time synchronization signal for SIUs. The wireless synchronization must provide clock synchronization accuracy of less than 4uS.	Demonstrate that the wireless clock synchronization accuracy is less than 4uS when no synchronization cable is connected between a base station and SIU (wireless synchronization active).
SIU Wireless Synchronization	Requirement ID: #8685wxruv AXON SIUs shall receive a wireless time synchronization signal from a Base Station. The wireless synchronization must provide clock synchronization accuracy of less than 4uS.	demonstrate that the wireless clock synchronization accuracy is less than 4uS when no synchronization cable is connected between a base station and SIU (wireless synchronization active).
Clock fault	Requirement ID: 868dt23nr The base station and associated software shall stop data transmission when the uncertainty of the base station clock and sampling system in SIU exceeds a predetermined threshold of no more than 50 microseconds (allowing the threshold to be adjusted).	Data transmission is stopped when uncertainty falls below a preconfigured threshold (no greater than 50 μ s).
Clock fault	Requirement ID: ID: 868dt22ym The base station and associated software shall indicate a fault when the uncertainty of the base station clock and sampling system in SIU exceeds a predetermined threshold of no more than 50 microseconds (allowing the threshold to be adjusted).	Fault is indicated when uncertainty falls below a preconfigured threshold (no greater than 50 μ s).

10. Revisions

Revision	Description	Prepared By	Date
A	Initial release	B. Timoszyk	DDMMYYYY Y
B	Added clock fault tests	M. Green	20Aug2025



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Create multiple datasheets to accommodate sample size as necessary.

Data Sheet		
DV Protocol:	EP-100161 Synchronization Accuracy Test DV Protocol	
Base Station Firmware Serial Number		
SIU Model		
SIU Firmware Serial Number		
Oscilloscope Manufacturer/Model		
Calibration Date/ Calibration Due		
Test Data		
Test Step	Data Description	Value
Synchronization	Synchronization accuracy (μ s)	
Clock fault	Clock drift after synchronization loss (μ s)	
Clock fault	AlertSyncFail count	
Data collected by:	Signature:	Date: